A Bayesian Treatment of the German Tank Problem: Conjugate Models, MCMC, Hierarchical Structure and Real-World Population Estimation

Project Proposal — Version 2

BSc Statistical Data Science, Year 3
Module: Bayesian Inference

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Abstract

The German Tank Problem asks for the size N of a population labelled $1, \ldots, N$ from a random sample of k labels drawn without replacement. This project treats it as a vehicle for the full Bayesian toolkit. We (i) derive exact posterior results and decision-theoretic estimates, (ii) study the choice of prior (conjugate, non-informative, Jeffreys-type, informative), (iii) use Bayes factors and posterior odds for decision problems such as "does the production exceed a legal cap?", (iv) build realistic sampling mechanisms from conjugate blocks (Beta–Bernoulli, Gamma–Poisson, Dirichlet–Multinomial, Gamma–Exponential, Normal–Gamma), (v) fit non-conjugate extensions with Gibbs and Metropolis–Hastings samplers, validated against exact posteriors, and (vi) embed everything in a hierarchical regression model with spike-and-slab variable selection, comparing hierarchical and empirical Bayes. Five applications are developed in depth: software defect registers, e-commerce order volumes, wildlife ringing, counterfeit detection, and NFT mint sizes.

1 Motivation and starting point

Let $S = \{s_1, \ldots, s_k\}$ be a uniform random subset of $\{1, \ldots, N\}$ and $m = \max S$. The frequentist solution is the UMVUE $\hat{N} = m(1 + 1/k) - 1$. An existing MSc dissertation on the topic (Gontla, QMUL, 2021) derives this, derives the flat-prior Bayesian posterior mean $\frac{(k-1)(m-1)}{k-2}$, and compares both by simulation at $N = 1000$, $k = 20$. This project goes further in the following ways:

- The dissertation's Poisson prior sets the rate μ to the average of the grid $\{0, \ldots, 2000\}$ and calls it a maximum-likelihood estimate; μ should instead be given a *hyperprior* (Gamma–Poisson) or estimated by empirical Bayes.
- Sensitivity of the posterior to the prior is reported as an error. In Bayesian analysis it is a feature to be *measured* (sensitivity analysis, Bayes-factor dependence on the prior), not eliminated.
- Methods are compared by the mean and standard deviation of point estimates only. Bayesian methods should be judged on the whole posterior: log score, interval score, and calibration of credible intervals.
- Only the idealised sampling mechanism is considered. Real data involve multiple series, selection effects, random sample sizes and time variation.

2 Syllabus map

Syllabus topic	Role in the project	Section
Exponential families, conjugate priors (Beta–Bernoulli, Gamma–Exponential, Gamma–Poisson, Normal–Gamma, Dirichlet–Multinomial)	Capture (thinning) model; Jelinski–Moranda detection model; prior on population size; second-stage model for log-intensities; factory shares	5
Bayesian estimation (posterior summaries, MAP, credible intervals)	Loss-function-based estimates; exact HPD intervals; comparison with UMVUE	4.2
Hypothesis testing (posterior odds, Bayes factor)	Legal-cap test; one versus several factories; selection bias present or not	4.4
Choice of priors (conjugate, non-informative, Jeffreys)	Flat, $1/n$, Pareto-type, Poisson/negative binomial, truncated; sensitivity analysis	4.3
Sampling from the posterior (Gibbs, Metropolis–Hastings, diagnostics)	Samplers for thinning and selection-bias models; trace/ACF/$\hat{R}$/ESS; validated against exact grids	6
Bayesian linear models (Normal-Inverse-Gamma, ridge/LASSO, spike-and-slab)	Regression of log-intensity on covariates; variable selection with posterior inclusion probabilities	7
Hierarchical models (hierarchical Bayes, empirical Bayes, profile likelihood)	Partial pooling across modules, months, sites, collections; HB versus EB; profile versus posterior intervals	7

3 Research questions

RQ1. Which point and interval estimates are optimal under which loss, and how do priors change them (and their frequentist coverage)?

RQ2. How strongly do Bayes factors and posterior odds for decision problems depend on the prior, and when are they trustworthy?

RQ3. Which realistic sampling mechanisms admit conjugate (closed-form) analysis, and where must MCMC take over?

RQ4. Does partial pooling (hierarchical Bayes) improve estimation when each unit has only a few observed serial numbers, and how does it compare with empirical Bayes?

RQ5. Which of the five applications actually satisfy the model's assumptions, and what is gained by modelling their violations?

4 Mathematical core

4.1 Likelihood and sufficiency

For $N \geq m$,

$$\mathbb{P}(S \mid N) = \binom{N}{k}^{-1}, \qquad \mathbb{P}(M = m \mid N) = \binom{m-1}{k-1}\binom{N}{k}^{-1}, \tag{1}$$

and 0 for $N < m$. The two differ by a constant free of N, so the pair (m, k) is sufficient and the posterior is a one-dimensional discrete distribution on $\{m, m+1, \dots\}$. This makes *exact* Bayesian computation possible in the base case and gives a ground truth against which every approximate method in the project is checked.

4.2 Bayesian estimation

For a prior $\pi(n)$ the posterior is $\pi(n \mid S) \propto \binom{n}{k}^{-1}\pi(n)\mathbf{1}\{n \geq m\}$. Under the improper flat prior $\pi(n) \propto 1$, using $\sum_{v \geq n}\binom{v}{k}^{-1} = \frac{k}{k-1}\binom{n-1}{k-1}^{-1}$, the posterior is proper for $k \geq 2$:

$$\pi(n \mid S) = \frac{k-1}{k}\binom{m-1}{k-1}\binom{n}{k}^{-1}, \qquad \mathbb{P}(N \geq n \mid S) = \binom{m-1}{k-1}\Big/\binom{n-1}{k-1}. \tag{2}$$

The posterior mean $\frac{(k-1)(m-1)}{k-2}$ is finite for $k \geq 3$ and the variance for $k \geq 4$ (the tail decays like n^{-k}).

Point estimates as Bayes rules. The estimate depends on the loss, which the application should dictate.

Loss	Bayes estimate	Comment
Squared error	posterior mean	compare with UMVUE
Absolute error	posterior median	robust to heavy tail
0–1 loss	MAP	equals m (the MLE) under flat prior: poor
Asymmetric linear, cost c_u per unit of underestimate, c_o per unit of overestimate	posterior τ-quantile, $\tau = \frac{c_u}{c_u+c_o}$	e.g. under-counting counterfeits costs more

Credible intervals. We compare equal-tailed and HPD intervals. Because the flat-prior posterior is decreasing on $n \geq m$, the HPD interval is $[m, u_\alpha]$ where u_α is the smallest n with $\binom{m-1}{k-1}/\binom{n}{k-1} \leq \alpha$. We examine frequentist coverage of these intervals by simulation, and contrast them with confidence intervals from the sampling distribution of m/N.

4.3 Choice of priors

Prior	$\pi(n)$	Type	Purpose
Flat	$\propto 1$ (improper)	non-informative	reproduces classical results
Log-uniform	$\propto 1/n$ (improper)	scale-invariant, Jeffreys-type	weakly informative on scale
Pareto-type	$\propto n^{-(\alpha+1)}$	conjugate in continuum limit	closed-form updating
Truncated uniform	$1/\Omega$ on $\{1, \ldots, \Omega\}$	bounded capacity	proper; needed for Bayes factors
Poisson(μ)	$e^{-\mu}\mu^n/n!$	informative	expected capacity
Gamma–Poisson	$\mu \sim \text{Gamma}(\alpha, \beta)$ gives $N \sim \text{NegBin}$	hierarchical	uncertainty about μ

Conjugacy in the continuum limit. For the continuous analogue $X_i \sim U(0, \theta)$ the Pareto(α, x_0) prior is conjugate: the posterior is Pareto($\alpha + k, \max(x_0, m)$). This family is *not* an exponential family (the support depends on θ), yet it has a one-dimensional sufficient statistic and a conjugate prior, which illustrates that conjugacy is not limited to exponential families. For the discrete problem, $\pi(n) \propto n^{-(\alpha+1)}$ gives a posterior approximately proportional to $n^{-(\alpha+1+k)}$ for large n.

Jeffreys-type prior. The Fisher regularity conditions fail for $U(0,\theta)$, so Jeffreys' rule can only be applied formally: with $E[\text{score}^2] = 1/\theta^2$ it gives $\pi(\theta) \propto 1/\theta$, which coincides with the scale-invariance argument. We examine what this prior implies for N and compare with the flat prior.

Sensitivity. For each prior we report posterior summaries and prior-predictive checks (does the prior imply plausible values of N and m?), and we vary the prior's hyperparameters systematically, including deliberately misspecified priors.

4.4 Hypothesis testing: posterior odds and Bayes factors

For competing models (or hypotheses) H_0, H_1 the Bayes factor is $\mathrm{BF}_{10} = p(S \mid H_1)/p(S \mid H_0)$, with marginal likelihood $p(S \mid H) = \sum_{n \geq m} \binom{n}{k}^{-1} \pi_H(n)$, and

$$\frac{\mathbb{P}(H_1 \mid S)}{\mathbb{P}(H_0 \mid S)} = \mathrm{BF}_{10} \times \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)}.$$

We study three types of test.

- **Legal-cap test.** $H_0 : N \leq L$ versus $H_1 : N > L$, equivalently $\mathbb{P}(N > L \mid S)$. If $m > L$ then $p(S \mid H_0) = 0$ and H_0 is excluded outright (up to violations of the model).
- **Point null.** $H_0 : N = N_0$, which has positive posterior mass because N is discrete.
- **Model comparison.** One versus two factories; no selection bias ($\theta = 0$) versus selection bias. For the nested case we use the Savage–Dickey ratio $\mathrm{BF}_{01} = \pi(\theta = 0 \mid S)/\pi(\theta = 0)$.

Two caveats are studied explicitly. Improper priors leave Bayes factors defined only up to an arbitrary constant, so proper (e.g. truncated) priors are required. Bayes factors are sensitive to the prior scale, which is the Lindley–Jeffreys phenomenon; we plot BF against prior width.

5 Conjugate building blocks for realistic sampling

5.1 Random sample size: Beta–Bernoulli and Gamma–Poisson

In practice k is not chosen by the analyst. Suppose $N \sim \mathrm{Pois}(\mu)$ and each unit is captured independently with probability p, with priors $p \sim \mathrm{Beta}(a, b)$ and $\mu \sim \mathrm{Gamma}(\alpha, \beta)$ (rate β). The likelihood of the captured set is $p^k(1-p)^{N-k}\mathbf{1}\{N \geq m\}$.

Proposition 1 (to be derived and checked numerically)**.** *The full conditionals are*

$$N \mid S, \mu, p \sim \mathrm{Pois}(\mu(1-p)) \text{ truncated to } n \geq m, \tag{3}$$

$$p \mid N, S \sim \mathrm{Beta}(a + k,\, b + N - k), \tag{4}$$

$$\mu \mid N \sim \mathrm{Gamma}(\alpha + N,\, \beta + 1). \tag{5}$$

Marginalising p gives $\pi(n \mid S, \mu) \propto \mathrm{Pois}(n; \mu)\, \mathrm{B}(a + k, b + n - k)$ for $n \geq m$, and marginalising μ gives the prior $N \sim$ NegBin with size α and success probability $\beta/(\beta+1)$.

All three conditionals are conjugate, so the Gibbs sampler of Section 6 needs no tuning. The pair (μ, p) is only weakly identified from (k, m), which is itself a lesson about priors: an informative $\mathrm{Beta}(a, b)$ prior on p (from domain knowledge or recapture data) matters.

5.2 Several series: Dirichlet–Multinomial

Let there be F factories (serial prefixes) with sizes N_f, total $N = \sum_f N_f$ and shares $\pi_f = N_f/N$. If the sample is uniform on the pooled population, the exact likelihood is

$$\mathbb{P}(S \mid N_{1:F}) = \binom{\sum_f N_f}{k}^{-1} \prod_f \mathbf{1}\{N_f \geq m_f\}. \tag{6}$$

When $k \ll N$ the counts $(k_1, \ldots, k_F)$ are approximately $\mathrm{Mult}(k, \pi)$, so with $\pi \sim \mathrm{Dir}(\alpha)$ the shares have posterior $\mathrm{Dir}(\alpha + k_{1:F})$, with a Dirichlet–Multinomial marginal likelihood. This fast conjugate approximation is used to initialise and cross-check the exact analysis, and its Bayes factor compares the one- and several-factory models.

5.3 Detection times: Gamma–Exponential (Jelinski–Moranda)

In the Jelinski–Moranda software reliability model, N initial defects each cause failure at rate ϕ, so the i-th inter-failure time is $t_i \sim \mathrm{Exp}(\phi(N-i+1))$. With $\phi \sim \mathrm{Gamma}(a, b)$ the rate integrates out in closed form:

$$p(t_{1:k} \mid N) = \frac{b^a\,\Gamma(a+k)}{\Gamma(a)}\,\frac{N!/(N-k)!}{(b+T_N)^{a+k}}, \qquad T_N = \sum_{i=1}^{k}(N-i+1)t_i, \tag{7}$$

so the exact posterior of N is again a one-dimensional sum under any prior on N. This is a second, independent information source about N to combine with serial-number data.

5.4 Second stage: Normal–Gamma

For log-intensities $y_j = \log \mu_j$ (production or defect rates across units) with $y_j \sim N(\eta, 1/\tau)$, $\eta \mid \tau \sim N(\eta_0, 1/(\kappa_0\tau))$, $\tau \sim \mathrm{Gamma}(a_0, b_0)$, the posterior is Normal–Gamma with

$$\kappa_J = \kappa_0 + J, \quad \eta_J = \frac{\kappa_0\eta_0 + J\bar{y}}{\kappa_J}, \quad a_J = a_0 + \tfrac{J}{2}, \quad b_J = b_0 + \tfrac{1}{2}\sum_j (y_j - \bar{y})^2 + \frac{\kappa_0 J(\bar{y}-\eta_0)^2}{2\kappa_J},$$

and a Student-t posterior predictive for a new unit.

6 Beyond conjugacy: MCMC

Gibbs sampler (thinning model). Cycle through the three conjugate conditionals of Proposition 1. A collapsed version (integrating out p) is compared for mixing.

Selection-bias model (Metropolis–Hastings). Suppose unit s is observed with weight $w(s;\theta) = s^{-\theta}$ (or $e^{-\theta s}$), with $\theta \sim N(0, \sigma_\theta^2)$. For $k \ll N$ the single-draw pmf is $w(s;\theta)/Z_N(\theta)$ with $Z_N(\theta) = \sum_{j \leq N} w(j;\theta)$, and

$$\pi(N, \theta \mid S) \;\propto\; \pi(N)\,\pi(\theta)\,\frac{\prod_{i=1}^{k} w(s_i;\theta)}{Z_N(\theta)^k}\,\mathbf{1}\{N \geq m\}. \tag{8}$$

The normaliser $Z_N(\theta)$ depends on N and must be kept; dropping it removes the penalty on large N. We sample N from its exact discrete full conditional and θ by random-walk Metropolis, tuning the step size to an acceptance rate of about 0.25–0.4. The exact without-replacement likelihood (a Wallenius-type form) is used as a check.

Diagnostics and validation.

- Graphical: trace plots, autocorrelation plots, rank plots, running means; numerical: split-$\hat{R}$ and effective sample size over several dispersed chains.
- Posterior predictive checks: simulate (k, m) from the posterior and compare with the observed value.
- **Exactness check:** in one or two dimensions (here N, or (N, θ)) the exact posterior is computable on a grid, so MCMC output is compared with it directly (total variation, KL).
- Simulation-based calibration of the whole inference pipeline.

7 Bayesian linear and hierarchical models

7.1 Full model

Units $j = 1, \ldots, J$ (software modules, months, sites or collections) each have a few observed labels. The model is

$$
\begin{aligned}
S_j \mid N_j, p_j &\text{ (thinned labels, as in Section 5)}, & p_j &\sim \mathrm{Beta}(a, b),\\
N_j \mid \mu_j &\sim \mathrm{Pois}(\mu_j), & y_j = \log \mu_j \mid \beta, \sigma^2 &\sim N(x_j^\top \beta, \sigma^2),\\
\beta_l \mid \gamma_l, \sigma^2 &\sim \gamma_l \, N(0, \sigma^2 \tau^2) + (1 - \gamma_l)\, \delta_0, & \gamma_l \mid \omega &\sim \mathrm{Bernoulli}(\omega),\\
\omega &\sim \mathrm{Beta}(a_\omega, b_\omega), & \sigma^2 &\sim \mathrm{InvGamma}(a_0, b_0).
\end{aligned}
$$

With $x_j \equiv 1$ the second stage reduces to the Normal–Gamma model above.

7.2 Bayesian linear models

Conditionally on the latent $y = (y_j)$, we have a linear Gaussian model. With a Normal–Inverse-Gamma prior $\beta \mid \sigma^2 \sim N(0, \sigma^2 V_0)$, $\sigma^2 \sim \mathrm{InvGamma}(a_0, b_0)$ the posterior is available in closed form: $V_J = (V_0^{-1} + X^\top X)^{-1}$, $\beta_J = V_J X^\top y$, $a_J = a_0 + J/2$, $b_J = b_0 + \frac{1}{2}(y^\top y - \beta_J^\top V_J^{-1} \beta_J)$, together with a closed-form marginal likelihood for Bayes-factor model comparison. We also study:

- **Ridge and LASSO as MAP estimates:** a Gaussian prior $\beta \sim N(0, \tau^2 I)$ gives ridge with $\lambda = \sigma^2/\tau^2$; a Laplace prior gives the LASSO. We compare these point estimates with the full posterior (and the Bayesian lasso Gibbs sampler if time permits).
- **Spike-and-slab variable selection:** the indicators γ are updated by Gibbs sampling using the collapsed Normal–Inverse-Gamma marginal likelihood; the Beta–Bernoulli structure on γ gives an automatic multiplicity correction. Output: posterior inclusion probabilities and posterior model probabilities, compared with the LASSO path.

7.3 Sampling the hierarchical model

Metropolis-within-Gibbs: N_j from a truncated Poisson, p_j from a Beta, y_j by random-walk Metropolis (the conditional $\propto \mu_j^{N_j} e^{-\mu_j} \times$ log-normal is not conjugate), $(\beta, \sigma^2, \gamma)$ by the conjugate/collapsed steps above, and ω from a Beta.

7.4 Hierarchical versus empirical Bayes; profile likelihood

- **Hierarchical Bayes:** hyperparameters (η, σ^2, ω, and so on) receive hyperpriors and are integrated out by MCMC.
- **Empirical Bayes:** hyperparameters are set to maximise the marginal likelihood (type-II maximum likelihood) and then plugged in. We study shrinkage towards the group mean for

units with small k_j, and show that plug-in intervals are too narrow because hyperparameter uncertainty is ignored.

- **Profile likelihood:** in the selection-bias model we profile out the nuisance parameter, $L_P(N) = \max_\theta L(N, \theta)$, and compare the likelihood-ratio interval with the posterior credible interval, especially when N and θ are weakly identified.

8 Applications in depth

For every application we state what N is, what the serial number is, which assumptions must hold, and which part of the Bayesian toolkit it exercises. The key validity condition is common to all of them: *the labels must be assigned independently of (and before) the mechanism that makes a unit observable.* If labels are assigned at the moment of observation (e.g. bug number i is the i-th bug reported), the labels are simply $1, \ldots, k$ and carry no information about N.

8.1 Software defect registers and testing

- **Quantity and labels.** N is the total number of defects or findings in a release or audit. Labels are informative when assigned at creation (pre-numbered test cases or static-analysis findings, seeded defects, or a tracker counter shared with hidden or private items), with a random subset reviewed.
- **Models.** (A) The thinning model with unit-level hierarchy over modules: μ_j regressed on code metrics (size, complexity, churn, number of authors) with spike-and-slab selection to identify which metrics matter. (B) The Jelinski–Moranda model for failure times, with closed-form marginal posterior for N.
- **Decisions.** A release rule such as $\mathbb{P}(N - k \leq r \mid \text{data}) \geq 0.95$; an asymmetric loss (cost of delay versus cost of a residual defect) gives a posterior quantile as the number of defects to plan for; a Bayes factor compares a constant per-defect rate with heterogeneous rates.
- **Data and validation.** Public defect datasets (NASA MDP/PROMISE-style repositories) and mined open-source trackers; truth is obtained by subsampling a fully known register and estimating from the subsample, so calibration can be measured.

8.2 E-commerce order volume

- **Quantity and labels.** N_t is the cumulative number of orders at date t; labels are sequential invoice or order numbers seen on receipts, screenshots or support tickets.
- **Production rate.** As in the original tank analysis, the quantity of interest is a *difference* $N_{t_2} - N_{t_1}$ from two snapshots. We model (N_{t_1}, N_{t_2}) jointly with the constraint $N_{t_2} \geq N_{t_1}$ and report the posterior of the difference, not the difference of two point estimates, which loses the strong dependence between them.
- **Hierarchy over periods.** Monthly log-rates have a Normal–Gamma second stage, giving partial pooling across months and a posterior predictive interval for next month.
- **Selection bias.** Support tickets over-represent recent and problematic orders: weights increasing in the order number, with a Savage–Dickey test of $\theta = 0$ and prior sensitivity of the estimate to the strength of the bias.
- **Data and ethics.** Order numbers may be randomised (UUIDs carry no information). Only data one is entitled to use are used: own-company data, public releases with sequential IDs, or a simulator calibrated to published statistics.

8.3 Wildlife ringing

- **What serial numbers estimate.** Sequential ring or tag numbers give the number of tags *deployed*, M (for example when ringing records are incomplete and rings are known only from recoveries). They do not estimate the animal population N_{pop} unless every animal is ringed, so this application is a two-stage model.
- **Stage 1.** German-tank posterior for M from recovered ring numbers.
- **Stage 2.** Mark–recapture: of n animals captured, r are marked, $r \sim \text{Bin}(n, p)$ with $p = M/N_{\text{pop}}$ (large-population approximation) and $p \sim \text{Beta}(a, b)$, so $p \mid r \sim \text{Beta}(a+r, b+n-r)$ and $N_{\text{pop}} = M/p$. The stages are composed by posterior sampling, so uncertainty in M propagates. The frequentist benchmark is the Chapman estimator $\hat{N} = \frac{(M+1)(n+1)}{r+1} - 1$.
- **Violations.** Ring loss and survival make older ring numbers less likely to be recovered, which is a selection weight in the labels. Heterogeneity across sites is handled hierarchically, with habitat covariates entering through spike-and-slab selection.

8.4 Supply chain and counterfeit detection

- **Quantity and labels.** N_f is the production run of factory (or licensed series) f with legal cap L_f; labels are printed serial numbers observed in the market, in seizures or in warranty claims.
- **Hypotheses.** H_0: all observed units come from licensed series with $N_f \leq L_f$; H_1: otherwise. A single serial above the cap rules out H_0 under the model, but it can also indicate a format change, so we report the Bayes factor and the posterior odds under several priors.
- **Contamination model.** Each observed unit is genuine with probability $1-\varepsilon$, with a serial uniform on the licensed range, or counterfeit with probability ε, with either a cloned serial (duplicates, a with-replacement signature) or a serial outside the range. Latent indicators and $\varepsilon \sim$ Beta give a Beta–Bernoulli Gibbs sampler.
- **Shares.** A Dirichlet–Multinomial model for the share of units by factory or distributor, with BF for one versus several factories.
- **Pitfalls.** Encrypted or non-sequential serials destroy the structure; seizures over-represent grey-market goods, which is a selection weight.

8.5 NFT and token mint sizes

- **A validation laboratory.** Token IDs on-chain are public and, for many collections, the true supply is readable from the contract. This gives real data with a known N: sample IDs as an API or marketplace would reveal them, estimate N, and measure calibration across many collections, which is naturally a hierarchical model over collections.
- **Selection.** Marketplaces and users favour rare traits; after a metadata reveal, rarity can correlate with ID, so $w(s; \theta)$ is non-constant. Partially revealed or non-contiguous ID ranges require the gap model.
- **Validity.** If IDs are assigned in mint order and observed during the mint, the max ID equals the count minted so far and carries no information about final supply; the model applies to ID assignment that is randomised relative to observation order.

9 Evaluation plan

- **Simulation grid:** $N \in \{100, 1000, 10^5\}$, $k \in \{4, 10, 20, 100\}$, with correct and misspecified priors and sampling mechanisms.

- **Metrics:** bias, RMSE, log score, CRPS, interval score, coverage of 95% credible intervals, ESS per second for samplers, total-variation distance of MCMC output from the exact posterior.
- **Benchmarks:** UMVUE, MLE, frequentist confidence intervals, Chapman estimator (wildlife), LASSO and ridge (variable selection), EB (versus HB).
- **Real data:** retrospective subsampling of registers with known N (software trackers, on-chain collections) and the WWII benchmark figures (statistical estimates 169, 244, 327 versus German records 122, 271, 342).

10 Optional extension: amortised neural posterior estimation

For extended mechanisms with intractable likelihoods, train a neural posterior estimator $q_\phi(N \mid S)$ on simulated (N, S) pairs with a permutation-invariant (DeepSets) encoder and a categorical output head. In the base case it is compared with the exact posterior, and elsewhere with MCMC and by simulation-based calibration. This is scoped as a stretch goal.

11 Deliverables and timeline

Deliverables: a report with verified derivations (flat-prior results, Proposition 1, Jelinski–Moranda marginal, hierarchical conditionals); a reproducible code repository (R and Python, own Gibbs and MH implementations validated against PyMC/NumPyro or Stan); and two full case studies plus two shorter ones.

Weeks	Tasks
1–2	Exact posterior, loss-based estimates, HPD, coverage; prior families and sensitivity; comparison with UMVUE
3	Bayes factors and posterior odds; prior dependence; Savage–Dickey
4–5	Conjugate blocks: thinning model, Dirichlet–Multinomial, Jelinski–Moranda, Normal–Gamma; numerical verification of the propositions
6–7	Gibbs and Metropolis–Hastings samplers; diagnostics; validation against exact posteriors; simulation-based calibration
8–9	Hierarchical regression with spike-and-slab; ridge/LASSO comparison; HB versus EB; profile likelihood
10–11	Applications (software and counterfeit in full, others lighter); optional neural estimator
12	Final report, code clean-up, presentation

12 Risks and mitigation

- **Weak identification** of (μ, p) or (N, θ) with small k: use informative priors, prior-predictive checks, and report the profile likelihood alongside the posterior.
- **Model validity** in applications: each case study starts with an explicit assumption checklist (are labels assigned before observation? sequential? independent of selection?) and stops if it fails.
- **Data access:** fall back on simulators calibrated to published figures.
- **Scope:** priority order is estimation, priors, Bayes factors, conjugate blocks, MCMC, hierarchy; the neural estimator and the time-varying model are optional.

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